DATA ANALYTICS FOR BUSINESS

with Lizhen Xu, Ph.D.

Course Introduction

Course Overview (1/3)

- **Course Expectation:**
  - Knowledge preparation: Statistics and math, algorithmic thinking, programming
  - Hands-on programming experience in R
  - Applications in various business contexts

- **Course Structure:**
  - One topic each week
  - Conceptual introduction followed by applications in R
  - Demonstration example in R for each topic
  - Homework exercises for each topic
    - Apply all key concepts for each topic to business applications in R
    - Important learning opportunities for this course

Course Overview (2/3)

- **Introduction**
  - Week 0: Introduction to R

- **Part 1: Econometric and Statistical Models**
  - Week 1: Linear Models
  - Week 2: Binary Response Models
  - Week 3: Censored/Truncated Data Models
  - Week 4: Count Data Models
  - Week 5: Survival Models
  - Week 6: Discrete Choice Models
  - Week 7: Causal Inference: Instrumental Variables

Non-linear Models
(Limited Dependent Variable Models)

Course Overview (3/3)

- **Part 2: Machine Learning Methods**
  - Week 8: Cluster Analysis
  - Collaborative Filtering
  - Week 9: Text Mining: Overview
  - Text Mining: Topic Modeling
  - Week 10: Neural Networks
  - Week 11: Deep Learning

Unsupervised Learning

Supervised Learning

References

- **Primary Textbook**
  - *An Introduction to Statistical Learning: with Applications in R* **(ISLR)**, 2nd Edition, by Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani (2021) Springer (ISBN 978-1071614174)

- **Supplementary Textbook**
  - *Machine Learning for Business Analytics: Concepts, Techniques, and Applications in R* **(MLBA)**, Second Edition, by Galit Shmueli, Peter C. Bruce, Peter Gedeck, Inbal Yahav, Nitin R. Patel (2023) John Wiley & Sons (ISBN 978-1119835172)
  - *Introductory Econometrics: A Modern Approach* **(Wooldridge)**, 7th Edition, by Jeffrey M. Wooldridge (2019) Cengage Learning (ISBN 978-1337558860)

- **Useful R Reference**
  - *The Book of R: A First Course in Programming and Statistics* by Tilman M. Davies (2016) No Starch Press (ISBN 978-1593276515)

DATA ANALYTICS FOR BUSINESS

with Lizhen Xu, Ph.D.

Overview of Data Analytics

Learning Objectives

- Explain:
  - Prediction vs. inference
  - Trade-off between prediction accuracy and model interpretability
- Explain:
  - Bias-variance trade-off
  - What is overfitting
- Explain the difference between:
  - Supervised and unsupervised learning
  - Regression and classification

What to Learn from Data?

$$Y = f(X) + \varepsilon$$

- Y: output, dependent (response/outcome) variables
- X: inputs, independent (explanatory/predicator) variables, features
- ε: error term (unobservable and unmeasurable variables)
- f: function, systematic dependence of Y on X, fixed but unknown

- **Data Analytics:**
  - A set of approaches for estimating f

$$\hat{Y} = \hat{f}\,(X)$$

Prediction vs Inference / Interpretation

- **Why estimate f**
  - Prediction
    - $\hat{f}$ is often treated as a ***black box***
  - Inference / Interpretation
    - $\hat{f}$ cannot be treated as a black box
    - Relationship between the response and the predictor variables

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning: with applications in R*. Springer. 2nd Ed. p 25. Retrieved March 29, 2022 from, https://hastie.su.domains/ISLR2/ISLRv2_website.pdf.

Prediction vs Inference / Interpretation (cont'd)

- **Prediction accuracy vs. Model interpretability**
  - *Trade-off---*
  - More *restrictive* models are *more interpretable*, but may have *less prediction accuracy*
  - More *flexible* models tend to have *higher prediction accuracy*, but are *less interpretable*

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning: with applications in R*. Springer. 2nd Ed. p 25. Retrieved March 29, 2022 from, https://hastie.su.domains/ISLR2/ISLRv2_website.pdf.

Bias-Variance Trade-off and Overfitting

- **Sources of Prediction Errors**

$$E(Y-\hat{Y})^2 = E\left[f(X)+\varepsilon-\hat{f}(X)\right]^2$$
$$= \underbrace{Var\left(\hat{f}(X)\right)+\left[Bias\left(\hat{f}(X)\right)\right]^2}_{\text{Reducible}} + \underbrace{Var(\varepsilon)}_{\text{Irreducible}}$$

- **Reducible vs Irreducible errors**
  - *Irreducible error*: Due to unobservable/unmeasurable variation in the data
  - *Reducible error*: The accuracy of $\hat{f}$ in making prediction
  - Goal: To minimize the reducible error

Bias-Variance Trade-off and Overfitting (cont'd)

- **Sources of Prediction Errors**

$$E(Y-\hat{Y})^2 = E\left[f(X)+\varepsilon-\hat{f}(X)\right]^2$$
$$= \underbrace{Var\left(\hat{f}(X)\right)+\left[Bias\left(\hat{f}(X)\right)\right]^2}_{\text{Reducible}} + \underbrace{Var(\varepsilon)}_{\text{Irreducible}}$$

- **Bias-Variance Trade-off**
  - ***Variance***: $Var\left(\hat{f}(X)\right)$, the amount by which $\hat{f}$ would change if we use a different *training* data set
  - ***Bias***: $Bias\left(\hat{f}(X)\right)$, the error from approximating the true f with a simpler model $\hat{f}$
  - Generally, more *flexible* models may have *lower bias*, but *higher variance*
  - Goal: To have *low variance* and *low bias*

- **Overfitting**
  - Follow the *noise* too closely, pick up patterns caused by *noise* rather than *systematic* relationship
  - Fit the training data well, but predict new test data poorly
  - A key issue to keep in mind when selecting machine learning methods

Categorization

- Supervised vs Unsupervised
  - ***Supervised***: has a response variable to predict (e.g., regression)
    - Explore the relationship between X variables and Y variable
  - ***Unsupervised***: has no response variable to predict (e.g., clustering)
    - Explore the data structure, relationship among X variables or among data observations

- Regression vs Classification
  - Regression: Numeric / quantitative response (continuous / discrete)
  - Classification: Categorical / qualitative response (nominal / ordinal)
  - *Note:* Independent variable types less important in determining ML methods
  - The boundary is blurry (e.g., logistic regression)

Summary

- Prediction vs. inference
- Trade-off: Prediction accuracy vs. model interpretability
- Bias-variance trade-off
- Overfitting
- Differences:
  - Supervised and unsupervised learning
  - Regression and classification